

Classic Riddler

1 April 2022

Riddle:

To play the game of “Zactoll” (named for yours truly, of course), you start with the number 1 and then try to reach a target number through a series of steps. For each step, you can *always* choose to double the number you currently have. However, if the number happens to be one more than an odd multiple of 3, you can choose to “reduce”—that is, subtract 1 and then divide by 3.

For example, to reach a target number of 5, you could double ($1 \rightarrow 2$), double ($2 \rightarrow 4$), double ($4 \rightarrow 8$), double ($8 \rightarrow 16$) and reduce ($16 \rightarrow 5$). And to get a target number of 6, you could double ($1 \rightarrow 2$), double ($2 \rightarrow 4$), double ($4 \rightarrow 8$), double ($8 \rightarrow 16$), reduce ($16 \rightarrow 5$), double ($5 \rightarrow 10$), reduce ($10 \rightarrow 3$) and double ($3 \rightarrow 6$).

What is the *smallest* target whole number you cannot reach in the game of Zactoll?

Solution:

This is an unsolved problem, because it is the inverse of the Collatz Conjecture. So the question has **no known solution**.